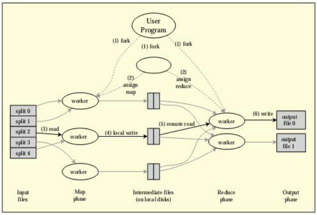

Figure 1: MapReduce: Simplified data processing on large clusters by Google

article, "whenever I mention Hadoop, I refer to the Hadoop Core package available from http://hadoop.apache.org".

There are five major components of Hadoop:

1. MapReduce (a job tracker and task tracker)
2. NameNode and Secondary NameNode
3. DataNode (that runs on a slave)
4. Job Tracker (runs on a master)
5. Task Tracker (runs on a slave)

MapReduce

The MapReduce framework has been introduced by Google. According to a definition in a Google paper on MapReduce, MapReduce is, “A simple and powerful interface that enables the automatic parallelisation and distribution of large-scale computations, combined with an implementation of this interface that achieves high performance on large clusters of commodity PCs.”

It has basically two components: Map and Reduce. The MapReduce component is used for data analytics programming. It completely hides the details of the system from the user.

HDFS

Hadoop has its own implementation of distributed file systems called Hadoop Distributed File System. It provides a set of commands just like the UNIX file and directory manipulation. One can also mount HDFS as fuse-dfs and use all the UNIX commands. The data block is generally 128 MB; hence, a 300 MB file will be split into 2 x 128 MB and 1 x 44 MB. All these split blocks will be copied ‘N’ times over clusters. N is the replication factor and is generally set to 3.

NameNode

NameNode contains information regarding the block’s location as well as the information of the entire directory structure and files. It is a single point of failure in the cluster, i.e., if NameNode goes down, the whole file system goes down. Hadoop therefore also contains a secondary NameNode which contains an edit log, which in case of the failure of NameNode, can be used to replay all the actions of the file system and thus restore the state of the

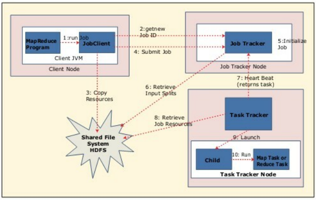

Figure 2: MapReduce behind the scenes

file system. A secondary NameNode regularly creates checkpoint images in the form of the edit log of NameNode.

DataNode

DataNode runs on all the slave machines and actually stores all the data of the cluster. DataNode periodically reports to NameNode with the list of blocks stored.

Job Tracker and Task Tracker

Job Tracker runs on the master node and Task Tracker runs on slave nodes. Each Task Tracker has multiple task-instances running, and every Task Tracker reports to Job Tracker in the form of a heart beat at regular intervals, which also carries details of the current job it is executing and is idle if it has finished executing. Job Tracker schedules jobs and takes care of failed ones by re-executing them on some other nodes. Job Tracker is currently a single point of failure in the Hadoop Cluster.

The overview of the system can be seen in Figure 2.

Hadoop in action

Let’s try a simple Hadoop word count example. First of all, you need to set up a Hadoop environment either in your own Linux environment or get a pre-configured virtual machine from *https://ccp.cloudera.com/display/SUPPORT/CDH+Downloads*. If you are willing to configure Hadoop by yourself, refer to the famous tutorial by Michael Noll, titled‘Running Hadoop on Ubuntu Linux (Multi-Node Cluster)’. Once you are done with the configuration of Hadoop, you can try out the example given below:

```
$HADOOP_HOME/bin/hadoop start-all.sh
jps
mkdir input
nano wordsIn \\fill this file with random words and exit or
copy a file into input dir in text format
$HADOOP_HOME/bin/hadoop dfs  -copyFromLocal  ./input  /user/
hduser/input
hadoop dfs  -cat  /user/hduser/input1/* \\ lists the content
```

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